Ring currents, local anisotropy, and the problem of the out-of-plane protons: a reinvestigation of the nuclear magnetic resonance spectrum of [10]-paracyclophane

ARVIND AGARWAL, JOHN A. BARNES, JOHN L. FLETCHER, MICHAEL J. MCGLINCHEY,[1] AND BRIAN G. SAYER

Department of Chemistry, McMaster University, Hamilton, Ont., Canada L8S 4M1

Received July 28, 1976[2]

ARVIND AGARWAL, JOHN A. BARNES, JOHN L. FLETCHER, MICHAEL J. MCGLINCHEY, and BRIAN G. SAYER. Can. J. Chem. **55**, 2575 (1977).

The ^{1}H nmr spectrum of [10]-paracyclophane has been recorded at 220 MHz and the methylene protons assigned on the basis of homonuclear decoupled spectra. Local anisotropic contributions to chemical shifts for protons sited above or near the aromatic ring of [10]-paracyclophane have been calculated using a classical model previously proposed by Grant. The residual incremental shift was shown to follow the Waugh–Fessenden–Johnson–Bovey classical ring current model, but the loop separation originally invoked was shown to be unnecessary. These corrected ring current contributions also correlate very well with the quantum-mechanical approach of Haigh and Mallion.

ARVIND AGARWAL, JOHN A. BARNES, JOHN L. FLETCHER, MICHAEL J. MCGLINCHEY et BRIAN G. SAYER. Can. J. Chem. **55**, 2575 (1977).

On a enregistré le spectre rmn du proton du [10]paracyclophane à 220 MHz et on a attribué les protons méthylènes en se basant sur le spectre découplé d'une façon homonucléaire. Des contributions anisotropes locales aux déplacements chimiques des protons situés au-dessus ou près du cycle aromatique du [10]paracyclophane ont été calculées en faisant appel à un modèle classique proposé antérieurement par Grant. On a montré que l'augmentation résiduelle du déplacement suit le modèle classique de courant de cycle de Waugh–Fessenden–Johnson–Bovey mais on a trouvé qu'il n'est pas nécessaire de faire appel à la séparation de boucle qui avait été invoquée originalement. Ces contributions de courant de cycle corrigées offrent aussi une très bonne corrélation avec l'approche de mécanique quantique de Haigh et Mallion.

[Traduit par le journal]

Introduction

The deshielding of aromatic protons relative to olefinic protons has occupied the attention of nmr spectroscopists and theoreticians for almost 20 years, and the collective wisdom of these deliberations has been the subject of a number of reviews (1–3). Following the ideas of Pauling (4), the ring current concept was applied to benzene by Pople (5) who treated the system as possessing a point dipole located at the centre of the ring and showed that this would account for the downfield shift of benzene relative to ethylene. In a now classic paper, Waugh and Fessenden (6) developed their "Free Electron Model" in which the π-electrons were regarded as being analogous to a simple current loop and, using the well-established methods of classical electrodynamics (7), they were able to evaluate the induced field at any point whether in or out of the plane of the ring. The downfield shift of the benzene protons predicted by an in-plane current loop was 2.79 ppm and, in order to correlate with the then accepted shift of ~1.5 ppm (*i.e.*, the difference between benzene and the vinylic protons of 1,3-cyclohexadiene), Waugh and Fessenden separated the π current into two loops, one above and one below the ring plane; the separation between the loops was *empirically* adjusted to be 1.28 Å, and a very convenient tabulation of chemical shift increments calculated on this basis has been provided by Johnson and Bovey (8). Following the quantum mechanical treatment of McWeeny (9), Haigh and Mallion (10, 11) have presented a similar tabulation; both Johnson and Bovey (8) and Haigh and Mallion (11) approaches allow the evaluation of chemical shifts for out-of-plane protons.

In their original paper, Waugh and Fessenden (6) applied their theory to out-of-plane protons and they presented experimental and calculated nmr shift data on [10]-paracyclophane. Although

[1]To whom all correspondence should be addressed.
[2]Revision received March 23, 1977.

Can. J. Chem. Downloaded from cdnsciencepub.com by 148.252.133.20 on 04/23/26

Can. J. Chem. Downloaded from cdnsciencepub.com by 148.252.133.20 on 04/23/26

their 40 MHz spectrum clearly showed the presence of high field protons in the methylene chain, the resolution was insufficient to allow any definitive assignments to be made. We now present the 220 MHz spectrum of [10]-paracyclophane (Fig. 1) in which the five aliphatic resonances are clearly resolved. These data allow a critical and comparative evaluation of both Johnson–Bovey (8) and Haigh–Mallion (11) ring current theories.

Experimental

Proton spectra were obtained on a Varian HR 220 spectrometer operating at 220 MHz and 18°C. Proton homonuclear decoupled spectra were obtained on a Varian HA 100 spectrometer operating at 100 MHz at 35°C using standard techniques.

Concentrations of 0.1–0.2 *M* were used in $CDCl_3$ as solvent, and the peak positions were extrapolated to infinite dilution; TMS was used as an internal standard.

A section of the 220 MHz spectrum is shown in Fig. 1, and the data are listed in Tables 1 and 2.

Results and Discussion

1. The ^1H Nuclear Magnetic Resonance Spectrum of [10]-Paracyclophane

As previously mentioned, a number of models have been proposed to account for the shifts of aromatic protons. The now widely accepted free electron model was used in an attempt to rationalize the shifts of the out-of-plane protons in [10]-paracyclophane; however, the lack of resolution in the 40 MHz spectrum made quantitative verification of the theory somewhat hazardous. Nevertheless, constant application to a variety of aromatic systems has almost bestowed an imprimatur upon the theory. However, with the technological advances of two decades, a number of discrepancies are now apparent. Thus, in 1957 there was no method of reliably determining the structure of [10]-paracyclophane (or at least the most populated conformation in solution) and the authors had to estimate the proton positions relative to the aromatic ring by using Fisher–Hirschfelder models and trying to allow for the twisting motions of the decamethylene bridge. It is perhaps not surprising that the original assignment of the methylene protons as following the sequence: α, β, δ, γ, ε is incorrect. The 220 MHz spectrum, together with the homonuclear decoupling data, unequivocably established the order of the proton resonances as: α, β, γ, δ, ε with the ε protons at highest field.

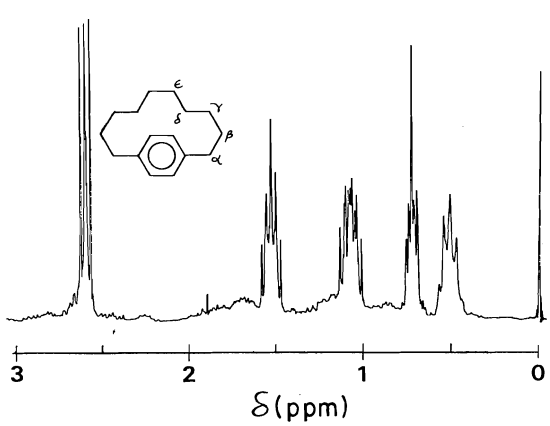

FIG. 1. 220 MHz ^1H nmr spectrum of high field resonances of [10]-paracyclophane.

TABLE 1. ^1H nmr data for [10]-paracyclophane

Position	δ values: Experimental 220 MHz	Calculated: Ref. 8 J–B	Calculated: Ref. 11 H–M
Ar	7.04		
α	2.62	1.66	1.60
β	1.54	1.55	1.36
γ	1.08	1.30	1.23
δ	0.73	0.82	1.13
ε	0.51	0.59	1.10

TABLE 2. Experimental vicinal coupling constants and calculated dihedral angles

Positions	*J* (Hz)	φ (Calculated, deg): Karplus	φ (Calculated, deg): Refined
Ring–α			155.5
α–β	6.2±0.1	62±3 or 112±3	62
β–γ	6.0±0.1	68±4 or 106±4	106
γ–δ	7.1±0.1	38±2 or 133±2	133
δ–ε	7.6±0.1	24±3 or 144±3	141.3

2. The Structure of [10]-Paracyclophane

The apparent inconsistency discussed above prompted us to direct our investigations toward a quantitative appraisal of the ring current tabulations (8, 11). Consequently, it was essential to know the structure of the molecule. In the absence of crystallographic data, it is necessary to calculate the molecular conformation of lowest energy, and for such systems a high degree of success has been achieved using Allinger's molecular mechanics calculations (12, 13). Thus, the correlation between the predicted and ob-

Can. J. Chem. Downloaded from cdnsciencepub.com by 148.252.133.20 on 04/23/26

served structures of [8]-paracyclophane is very good and so one can confidently use this approach as the best available at present. Since the ^{1}H nmr spectrum shows only five methylene environments, the molecule is conformationally mobile on the nmr time scale; indeed Allinger calculated that the molecule would have a rather flat potential energy minimum allowing considerable amplitude of ring oscillation. The preferred conformation for [10]-paracyclophane was calculated to have a non-planar aromatic ring in which the *para*-carbons were displaced upwards through 8.4°, while the α-carbons were displaced downwards through 4.4° (13). The molecular mechanics method also predicted the conformation of the decamethylene chain, and an attempt to verify these predictions was undertaken using the Karplus theory (14, 15) relating vicinal coupling constants ($^3J_{HH}$) and dihedral angles which has been extensively used in structure determinations (16).

The dihedral angles of the most populated conformation were determined from the coupling constants obtained from the homonuclear decoupled spectra using the following relationship:

$$[1] \qquad J = A + B\cos\phi + C\cos 2\phi$$

where $A = 7$ Hz, $B = -1$ Hz, $C = 5$ Hz, a set of empirically derived parameters which refer to an H—C—C—H fragment in a carbon chain (17). The observed coupling constant is in fact an average of four coupling constants whose dihedral angles of interaction are ϕ, $120 + \phi$, ϕ, and $|120 - \phi|$. Then,

$$[2] \quad J_{obs} = 0.25(2J_{\phi} + J_{\phi+120} + J_{120-\phi})$$

with this consideration, eq. 1 becomes:

$$[3] \quad J_{obs} = 2.5\cos^2\phi - 0.25\cos\phi + 5.75$$

The calculated dihedral angles are listed in Table 2. In this analysis it was assumed that each —CH_2—CH_2— unit could be regarded as an $[AX]_2$ system, since the chemical shifts of the protons in a given methylene group are dynamically averaged to give only one value and the chemical shift differences between adjacent methylene groups are relatively large. While this simplification is justifiable for the $J_{\alpha\beta}$, $J_{\beta\gamma}$, and $J_{\gamma\delta}$ couplings such is not the case for the $J_{\delta\varepsilon}$ case which is more correctly regarded as $[[AB]_2]_2$; consequently the dihedral angle predictions from $^3J_{H_\delta H_\varepsilon}$ are distinctly less reliable than are the others. Of course, the dihedral angle between the aromatic ring and the α-carbon—β-carbon bond is not evaluable directly as there is no appropriate coupling constant. This angle was approximated from a Fieser model and refined as outlined below. Now, accepting the values for the non-planarity of the ring and C_α atoms as calculated by Allinger (13), use of molecular models made it relatively easy to decide the unique combination of dihedral angles which allowed the construction of the decamethylene bridge. Finally, a molecular geometry programme (18) was used empirically to adjust not only the dihedral angles (within the calculated error limits) but also the aliphatic C—C—C bond angles (to parallel the behaviour of the corresponding angles in [8]-paracyclophane) such that the ε-carbon coordinates (x,y,z) corresponded to the ε′-carbon coordinates $(\bar{x},\bar{y},z)$ as required by C_2 symmetry. Throughout these calculations, the aromatic ring carbon–carbon bond lengths were taken as 1.39 Å, the exocyclic sp^2–sp^3 bond as 1.50 Å and the C—C and C—H bonds in the polymethylene chain as 1.54 and 1.08 Å respectively. Each H—C—H plane was positioned orthogonally to the appropriate C—C—C plane. It is gratifying that these experimental data, *viz.*, coupling constants, reinforce Allinger's theoretically predicted molecular geometry as shown by comparing Fig. 2 with Fig. 2 in ref. 13.

3. *Comparison of Ring Current Theories*

These structural data allow the evaluation of incremental shifts for the methylene protons lying out of the aromatic ring plane using both the semiclassical (8) and quantum mechanical (11) approaches. In order to predict chemical shifts relative to TMS it is necessary to select the "standard methylene shift" appropriate to the system. Typical values for large cycloalkanes average to 1.30 ppm (19), and this is the standard we have used. The chemical shifts thus predicted by the Johnson–Bovey (8) and Haigh–Mallion (11) tabulations for the α, β, γ, δ, and ε protons are presented in Table 1, and may be compared with the experimental values in the same table. It is apparent that, while both methods are qualitatively very useful, the quantitative agreement with the experimental shifts is not uniform throughout all positions which experience the effect of the ring current. The semi-classical method shows good agreement for protons held

Can. J. Chem. Downloaded from cdnsciencepub.com by 148.252.133.20 on 04/23/26

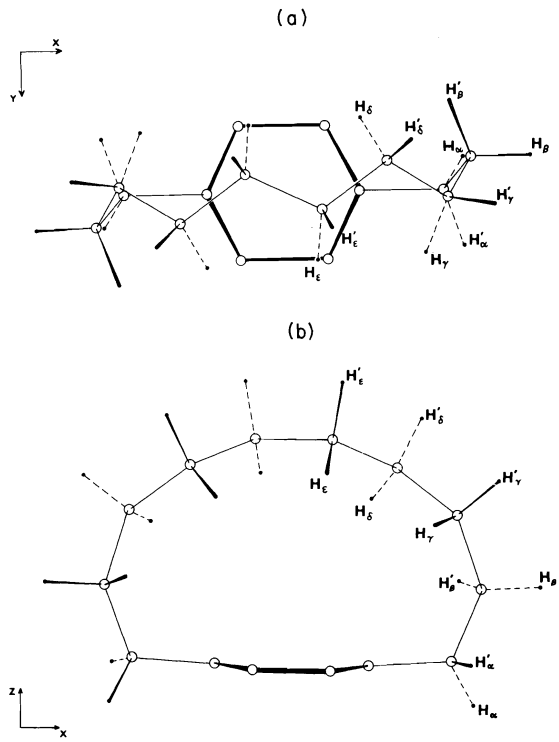

FIG. 2. Preferred conformations of [10]-paracyclophane: (*a*) plan, (*b*) elevation, the hydrogen atom positions are shown in perspective relative to the carbon atom to which they are attached.

over the centre of the ring, but becomes less reliable as the value of ρ increases and *z* decreases. However, the quantum mechanical approach gives adequate agreement in these latter regions but seriously underestimates the upfield shifts of the protons directly above the ring, as was originally inferred (10, 20). The work of Rose (21) on pyridylphenanthrenes also supports this generalization.

4. *Local Anisotropic Contributions*

It was apparent that the present ring current theories did not provide a complete rationale for the experimentally observed shifts and that some other important contribution had been overlooked. It had earlier been suggested by Pople (22, 23) that local anisotropic effects were a large contributing factor to the deshielding of the protons in benzene. Furthermore Barfield, Grant, and Ikenberry (24) have now successfully demonstrated that local anisotropic contributions to chemical shifts may be evaluated using the ^{13}C chemical shielding tensors so painstakingly obtained by Waugh and co-workers (25, 26) using proton-enhanced nuclear induction spectroscopy (27). Grant's data (24) clearly show that local anisotropic contributions cannot be ignored for in-plane protons in condensed aromatic systems,[3] and we now extend these calculations to out-of-plane protons.

Using the analogy of a current circulating in a loop of wire (24), the proton shielding component, σ_{ii}^{H}, is given by

$$[4] \quad \sigma_{ii}^{H} = \frac{a}{\pi} \frac{1}{[(a+\rho)^2 + z^2]^{1/2}} \times \left[K + \frac{a^2 - \rho^2 - z^2}{(a-\rho)^2 + z^2} \cdot E \right] \cdot \sigma_{ii}^{C}$$

where *a* is the radius of circulation about each carbon atom, taken as 0.47 Å (24), ρ and *z* are the usual cylindrical coordinates, and *K* and *E* are complete elliptical integrals of the first and second type with the argument

$$k^2 = \frac{4a\rho}{(a+\rho)^2 + z^2}$$

More simply, $\sigma_{ii}^{H} = G\sigma_{ii}^{C}$ where σ_{ii}^{H} and σ_{ii}^{C} are the proton and carbon shielding components and *G* is a geometric factor. The isotropic shielding, as measured in solution, is given by averaging the components

$$\langle \sigma^{H} \rangle = \tfrac{1}{3}[\sigma_{11}^{H} + \sigma_{22}^{H} + \sigma_{33}^{H}]$$

The ^{13}C shielding tensors used in this study are taken from refs. 25 and 26; values used for aromatic carbons bearing alkyl chains are those listed by Waugh for hexaethylbenzene, while those for aromatic carbons with hydrogen substituents are the average of the appropriate values from pentamethylbenzene and durene. To convert the values from shifts relative to benzene into absolute shifts, it is necessary to add 71 ppm to each ^{13}C shielding tensor (24). The local anisotropic contributions for each methylene proton brought about by circulations about each ring carbon in [10]-paracyclophane are presented in Table 3.

[3]A dipolar approximation presented in ref. 24 contains an algebraic error due to mis-evaluation of the elliptical integral power series. However, this does not materially affect the conclusions drawn previously, *i.e.*, it is unsafe to calculate chemical shifts for proximate protons using only a dipolar formula. For the situation in ref. 24, the correct dipolar formula is

$$\langle \sigma^{H} \rangle = \frac{a^2}{3r_{CH}^2} \left(\sigma_{11}^{C} + \frac{a}{r_{CH}} \cdot \sigma_{22}^{C} + \sigma_{33}^{C} \right)$$

Can. J. Chem. Downloaded from cdnsciencepub.com by 148.252.133.20 on 04/23/26

5. Experimentally Determined Ring Currents

We can now define the ring current contribution (ΔRC) to the chemical shift as the difference between the experimental shift (δ) and that calculated by adding the local anisotropic contribution to the standard, *i.e.*,

$$[5] \quad (-\delta)_{\text{expt}} = -1.30 + \text{LA term} + \Delta\text{RC}_{\text{expt}}$$

These "experimentally determined ring current shifts" are listed in Table 3 together with ring current shifts calculated using the equation

$$[6] \quad \Delta = \frac{C}{[(a+\rho)^2 + z^2]^{1/2}} \times \left[K + \frac{a^2 - \rho^2 - z^2}{(a-\rho)^2 + z^2} \cdot E\right]$$

where $C = ne^2/6\pi M_e c^2 = 8.970$, where n is the number of circulating electrons of charge e and mass M_e, c is the velocity of light, ρ and z are the usual cylindrical coordinates, and a is the radius of circulation about the ring taken as 1.39 Å. In this case ρ and z are measured in Å rather than ring radii.

The quoted shifts are the average of two shifts Δ_1 and Δ_2 obtained by taking $z_1 = z + S/2$ and $z_2 = z - S/2$, where S is the separation of the current loops. It is necessary to accept that any ring current contribution calculated semi-classically must follow an equation of the type [6] where the only adjustable parameters are C and S. Waugh and Fessenden chose to adjust the separation of the loops, while Barfield, Grant, and Ikenberry adjusted both C and S. Our approach was to adjust the separation between the loops to get the best correlation between the "experimental ring current shifts" and the calculated shifts, and then to evaluate C empirically.

It is now apparent from our data that the arbitrary separation into two loops is not only unnecessary but is actually detrimental to the optimum fit. Now a plot (Fig. 3) of the $\Delta\text{RC}_{\text{expt}}$ values *vs.* calculated ring current by the Johnson–Bovey (8) method ($\Delta\text{RC}_{\text{J-B},S=0}$) provides a value for C, and follows the equation

$$[7] \quad \Delta\text{RC}_{\text{expt}} = 1.04\,\Delta\text{RC}_{\text{J-B},S=0} - 0.06$$

The correlation coefficient, $\bar{r}$, was found to be a maximum (0.9928) at $S = 0$; the correlation worsened very rapidly on invoking loop separation. Typically, $S = 1.0$ Å, $\bar{r} = 0.9461$; $S = 1.28$ Å (ref. 6), $\bar{r} = 0.8883$; $S = 1.63$ Å (ref. 24),

TABLE 3. Calculated and experimental ring current and local anisotropy incremental contributions to the shifts of protons in the decamethylene bridge of [10]-paracyclophane. H′ represents the proton of any given methylene pair having the larger positive z value

	Ring current increments[a]							
	Semi-classical theory				Quantum mechanical theory			Local anisotropy increments
Atom	$S = 0$		$S = 1.28$[b]	$S = 1.63$[c]			Experimental	
H_α	−0.729	−0.786	−0.379	−0.228	−0.278	−0.300	−0.905	−0.295
H_α'	−0.842		−0.340	−0.178	−0.321			−0.535
H_β	−0.173	−0.171	−0.171	−0.161	−0.069	−0.063	−0.155	−0.104
H_β'	−0.169		−0.332	−0.351	−0.057			−0.067
H_γ	0.265	0.177	−0.030	−0.248	0.102	0.068	0.095	0.193
H_γ'	0.088		0.026	−0.015	0.034			0.057
H_δ	0.643	0.446	0.692	0.652	0.246	0.171	0.329	0.348
H_δ'	0.248		0.266	0.271	0.095			0.134
H_ε	0.717	0.515	1.014	1.232	0.274	0.197	0.538	0.347
H_ε'	0.313		0.405	0.472	0.120			0.157

[a]Units: ppm. A minus sign indicates a downfield effect.
[b]Johnson–Bovey separation (Å), ref. 8.
[c]Barfield–Grant–Ikenberry separation (Å), ref. 24.

Can. J. Chem. Downloaded from cdnsciencepub.com by 148.252.133.20 on 04/23/26

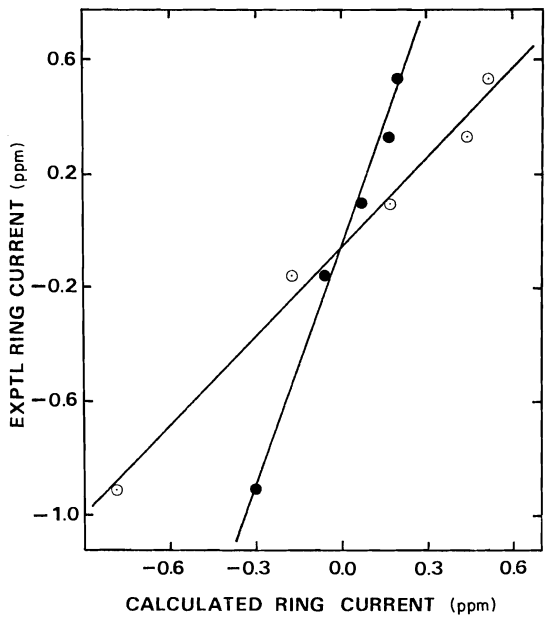

FIG. 3. Plot of experimental ring current (ΔRC_{expt}) *vs.* calculated ring current ($\Delta RC_{J-B,S=0}$, ○; and ΔRC_{H-M}, ●).

$\bar{r} = 0.7634$; $S = 2.0$ Å, $\bar{r} = 0.6161$. Therefore, the data force upon us the conclusion that the arbitary separation of the ring current into two loops is unnecessary; furthermore, the ring current increments derived by a semi-classical approach and those determined experimentally show good agreement.

A plot of measured chemical shift ($-\delta$) *vs.* calculated ring current ($\Delta RC_{J-B,S=0}$) also shows an interesting correlation, *viz.*,

$$[8] \quad (-\delta) = 1.57\,\Delta RC_{J-B,S=0} - 1.36 \qquad (\bar{r} = 0.9974)$$

Combining eqs. 5, 7, and 8, we see that

$$[9] \qquad LA = 0.53\,\Delta RC_{J-B,S=0}$$

indicating that the ring current contribution to the incremental shift is of the order of 65%.

This whole postulate receives strong support from Dailey's work on magnetic anisotropy which showed that the ring current only contributes to the extent of ~60% to the chemical shifts of aromatic hydrocarbons (28).

Having shown that the experimental ring current shifts are readily rationalizable from the semi-classical viewpoint, it is of interest to determine whether any analogous correlation exists for the quantum mechanical theory. This latter theory has only one variable parameter which was obtained from a linear regression analysis of data for planar polycyclic aromatics.

A plot of our experimental ring current shifts against those predicted from the quantum mechanical approach shows excellent agreement ($\bar{r} = 0.9932$) and follows the equation

$$[10] \qquad \Delta RC_{expt} = 2.73\,\Delta RC_{H-M} - 0.06$$

It will be noted that the predictions of the Johnson–Bovey (8) and Haigh–Mallion (11) tabulations differ by a factor of ~2.6 in the limit (11). Using our data (see Table 3) the following relationships were found

$$[11] \qquad \Delta RC_{J-B,S=0} = 2.60\,\Delta RC_{H-M}$$

$$[12] \qquad \Delta RC_{J-B,S=1.28} = 2.02\,\Delta RC_{H-M}$$

where $\bar{r} = 0.9998$ and 0.8938 respectively, thus reinforcing our earlier arguments dispelling the necessity for loop separation.

It is especially gratifying to see that the two entirely different models are compatible with the experimental data. Previous statements (11, 21) that the two theories are only applicable to specific spatial regions of the aromatic molecule are nullified when the correct ring current increments are evaluated. Thus previously published experimental ring currents which did not allow for the local anisotropic corrections would not be expected to fit either ring current model, except fortuitously.

One would have anticipated that eqs. 7 and 10 should have zero intercept since the overall chemical shift of any given methylenic proton is assumed to be a composite of only three terms *viz.*, LA and RC terms, which are of course position dependent, and the standard shift, which is the same for all protons in the chain. Since one obtains the identical non-zero intercept from both the semi-classical and quantum mechanical approaches, we propose that the standard methylene shift in chloroform be taken as 1.36 ppm rather than the initially chosen value of 1.30 ppm.

In the previous treatment of [10]-paracyclophane (6), the predicted shifts of the protons α to the ring gave such poor agreement that they were ignored, and these shifts were merely compared to the corresponding protons in ethylbenzene. However, the present approach adequately accounts for the α-CH_2 shift *without* invoking an inductive effect of the phenyl ring; that is, the inherent standard shift for the α-proton can still

Can. J. Chem. Downloaded from cdnsciencepub.com by 148.252.133.20 on 04/23/26

be taken as 1.36 ppm. Thus, it would seem that the so-called inductive effect of the phenyl ring can be wholely attributed as being a manifestation of the anisotropic nature of the electron distribution within the ring.

In calculating incremental methylene proton shifts using a combination of local anisotropic and ring current contributions *from the aromatic ring*, we have assumed that local anisotropic contributions from atomic neighbours in the polymethylene chain are allowed for by suitable choice of a standard. One should point out[4] that whilst our determination of the local currents involves the use of the total carbon shielding tensor (the effect of the delocalised current being ignored), comparison of the isotropic ^{13}C shifts in benzene and for the C-2, C-3 resonances in cycloheptatriene shows that the delocalised effect must be relatively small.

We feel that this approach to the problem of the ring currents is valuable in that the investigation of the methylene protons is less hazardous than investigating the aromatic protons for which the choice of a standard shift is more difficult and controversial. An extension of the concepts developed here allows a study of ring currents in arene-metal π-complexes and this will be dealt with in a subsequent publication.

Conclusions

Present ring current tabulations do not allow for the local anisotropic contributions to the chemical shifts of protons in aromatic environments; independent calculation of these local anisotropic contributions allows the evaluation of corrected (experimental) ring current shifts. These corrected shifts were found to show simple relationships to both the quantum mechanical and semi-classical approaches; in each case, the correlation coefficients were excellent. However, the separation of the current loops in the semi-classical approach while aesthetically satisfying was shown to be unnecessary.

Acknowledgements

We wish to express our sincere gratitude to Professor Norman L. Allinger (University of Georgia) for his most generous gift of the sample of [10]-paracyclophane and for discussions on molecular geometry. We also wish to thank Dr. A. A. Grey for use of the 220 MHz spectrometer facilities (Ontario Research Foundation, Sheridan Park Research Centre, Mississauga, Ontario), Dr. F. G. Riddell (University of Stirling, Scotland) for helpful discussions, and our colleague Professor D. P. Santry for computational advice.

Financial assistance from the National Research Council of Canada is gratefully acknowledged.

[4]We thank a particularly diligent referee for this and other valuable comments on the manuscript.

1. A. J. Jones. Rev. Pure Appl. Chem. **18**, 253 (1968).
2. R. C. Haddon, V. R. Haddon, and L. M. Jackman. Top. Curr. Chem. **16**, 103 (1971).
3. J-F. Labarre and F. Crasnier. Top. Curr. Chem. **24**, 33 (1971).
4. L. Pauling. J. Chem. Phys. **4**, 673 (1936).
5. J. A. Pople. J. Chem. Phys. **24**, 1111 (1956).
6. J. S. Waugh and R. W. Fessenden. J. Am. Chem. Soc. **79**, 846 (1957); **80**, 6697 (1958).
7. W. R. Smythe. Static and dynamic electricity. 3rd ed. McGraw-Hill, New York, NY. 1968. pp. 290, 291.
8. C. E. Johnson, Jr. and F. A. Bovey. J. Chem. Phys. **29**, 1012 (1958).
9. R. McWeeny. Mol. Phys. **1**, 311 (1958).
10. C. W. Haigh and R. B. Mallion. Mol. Phys. **22**, 995 (1971).
11. C. W. Haigh and R. B. Mallion. Org. Magn. Reson. **4**, 203 (1972).
12. N. L. Allinger and J. T. Sprague. J. Am. Chem. Soc. **95**, 3893 (1973).
13. N. L. Allinger, J. T. Sprague, and T. Liljefors. J. Am. Chem. Soc. **96**, 5100 (1974).
14. M. Karplus. J. Chem. Phys. **30**, 11 (1959).
15. M. Karplus. J. Am. Chem. Soc. **85**, 2870 (1963).
16. H. Booth. *In* Progress in nuclear magnetic resonances spectroscopy. Vol. 5. *Edited by* J. W. Emsley, J. Feeney, and L. H. Sutcliffe. Pergamon Press, Oxford. 1969. p. 149.
17. A. A. Bothner-By. Adv. Magn. Reson. **1**, 195 (1965).
18. P. E. Stevenson and J. E. Merrill. Local version of COORD, Atomic Cartesian Coordinates for Molecules, Program 186, Quantum Chemistry Program Exchange, Indiana University, 1971.
19. J. J. Burke and P. C. Lauterbur. J. Am. Chem. Soc. **86**, 1870 (1964).
20. C. W. Haigh. Personal communication.
21. P. I. Rose. Org. Magn. Res. **5**, 187 (1973).
22. J. A. Pople. J. Chem. Phys. **41**, 2559 (1964).
23. A. F. Ferguson and J. A. Pople. J. Chem. Phys. **42**, 1560 (1965).
24. M. Barfield, D. M. Grant, and D. Ikenberry. J. Am. Chem. Soc. **97**, 6956 (1975).
25. S. Pausak, A. Pines, and J. S. Waugh. J. Chem. Phys. **59**, 591 (1973).
26. S. Pausak, J. Tegenfeldt, and J. S. Waugh. J. Chem. Phys. **61**, 1338 (1974).
27. A. Pines, M. G. Gibby, and J. S. Waugh. Chem. Phys. Lett. **15**, 373 (1972).
28. B. P. Dailey. J. Chem. Phys. **41**, 2304 (1964).